



Development of a 3D Model and CNC Machining Program Using G-code

Prototyping of Structures Using 3D Printing and CNC Techniques

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15.05.2026

1. Aim and Description of the Task

The aim of the project was to prepare a complete CNC machining program for a part designed in a CAD/CAM environment and to generate G-code that allows the part to be machined on a CNC milling machine.

The project included:

- preparing the 3D model,
- selecting tools,
- setting the machining parameters,
- performing a facing operation,
- pocket milling,
- contour machining,
- drilling using the G81 cycle,
- generating and analysing the G-code.

The part has a milled pocket, an outer contour and a hole made by a drilling operation.

2. Theoretical Background

CNC machining consists of controlling the motion of the tool by means of a numerical program written as G-code. The program defines the tool movements, the cutting parameters and the auxiliary functions that control the machine tool.

The most important commands used in the project:

Command	Meaning
G0	Rapid move
G1	Linear interpolation
G2/G3	Circular interpolation
G17/G18	Plane selection
G21	Units in mm
G43	Tool length compensation
G54	Workpiece coordinate system
G81	Drilling cycle
M3	Spindle start
M5	Spindle stop
M6	Tool change
M8/M9	Coolant ON/OFF
M30	End of program

Formulas used in machining:

Cutting speed

$$V_c = \frac{\pi \cdot D \cdot n}{1000}$$

where:

- V_c – cutting speed [m/min],
- D – tool diameter [mm],
- n – spindle speed [rpm].

Feed per tooth

$$f_z = \frac{F}{z \cdot n}$$

where:

- f_z – feed per tooth [mm/tooth],
- F – feed rate [mm/min],
- z – number of teeth (flutes),
- n – spindle speed [rpm].

3. Description of the Part and Material

The designed part has the form of a rectangular block with the following dimensions:

- length: 100 mm,
- width: 80 mm,
- height (extrusion): 30 mm.

A milled pocket 20 mm deep was made in the central part of the block. In addition, the outer contour was machined and a hole was made by a drilling operation.

The workpiece material was steel. It was chosen because it allows full milling and drilling to be carried out while keeping the part sufficiently rigid.

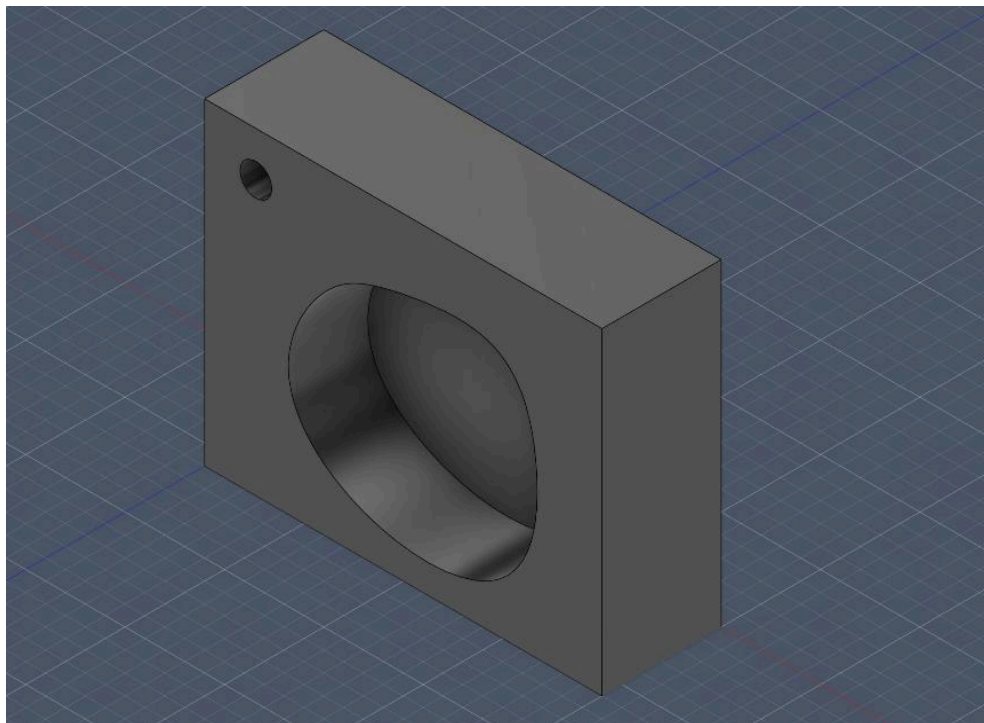


Fig. 1: 3D model of the designed part

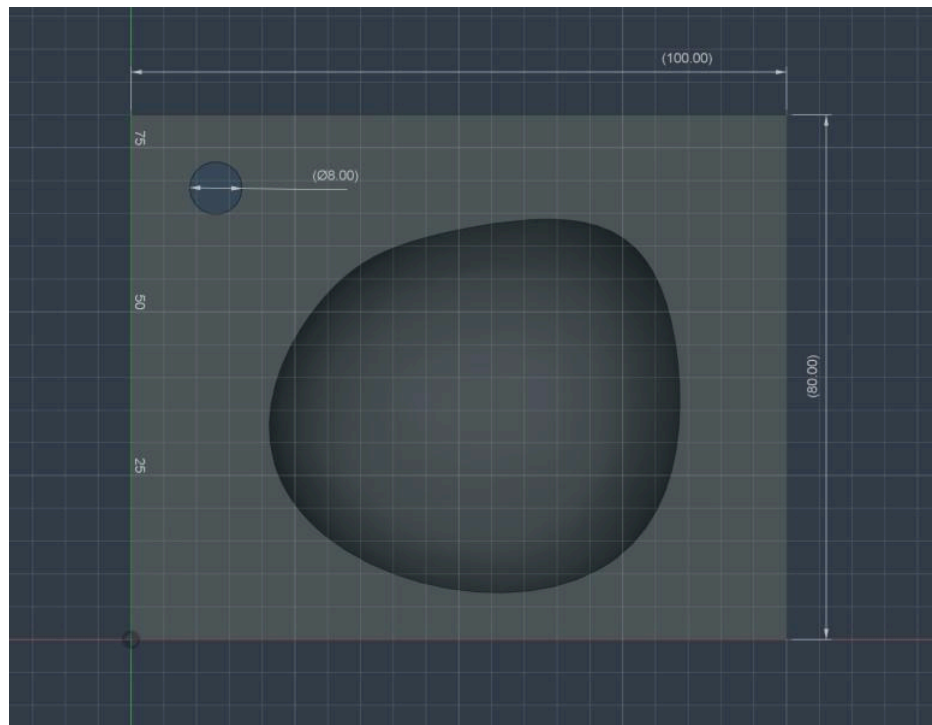


Fig. 2: Dimensioned sketch of the face

4. Tool Selection

Three tools were used in the project, for facing, milling and drilling.

Tool	Type	Diameter	Application
T1	Flat end mill	10 mm	Face milling
T2	Flat end mill	6 mm	Pocket and contour milling
T3	118° twist drill	8 mm	Drilling the hole

The tool definitions are at the beginning of the program:

```
(T1 D=10. CR=0. - ZMIN=-1. - FLAT END MILL)
(T2 D=6. CR=0. - ZMIN=-31. - FLAT END MILL)
(T3 D=8. CR=0. TAPER=118DEG - ZMIN=-31. - DRILL)
```

The Ø10 mm end mill was used for facing because it allows material to be removed quickly. The Ø6 mm end mill was used for the more precise machining of the pocket and the contours. The hole was made with a Ø8 mm drill with a 118° point angle.

5. Cutting Parameters and Calculations

The program uses:

- spindle speed: S5000,
- cutting feed rate: F1000,
- plunge and drilling feed rate: F333.33.

Cutting speed for T1 (Ø10 mm)

$$V_c = \frac{3.14 \cdot 10 \cdot 5000}{1000} \approx 157 \text{ m/min}$$

Cutting speed for T2 (Ø6 mm)

$$V_c = \frac{3.14 \cdot 6 \cdot 5000}{1000} \approx 94.2 \text{ m/min}$$

Feed per tooth

For a two-flute end mill:

$$f_z = \frac{1000}{2 \cdot 5000} = 0.1 \text{ mm/tooth}$$

The parameters used are suitable for machining steel and provide stable tool operation and good surface quality.

6. Project Workflow

First, the 3D model of the part and the stock were prepared. Then the workpiece coordinate system G54 was set.

The following operations were prepared in Autodesk Fusion, in this order:

1. Face milling,
2. Pocket milling,
3. Contour machining,
4. Drilling.

A suitable tool and machining parameters were chosen for each operation.

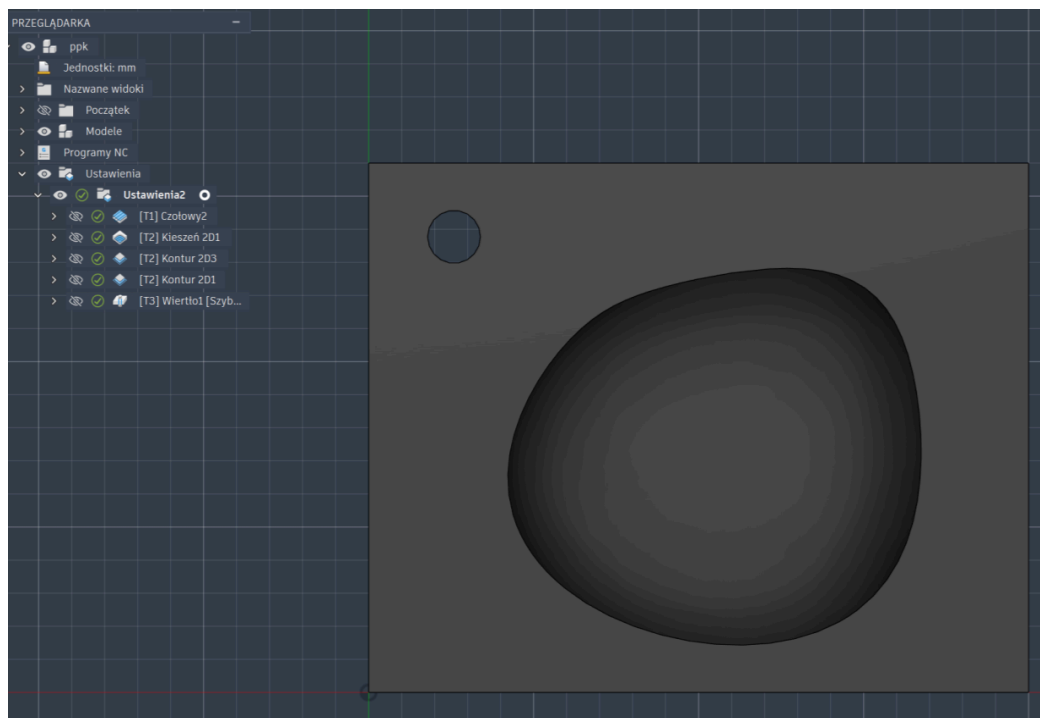


Fig. 3: Facing toolpath

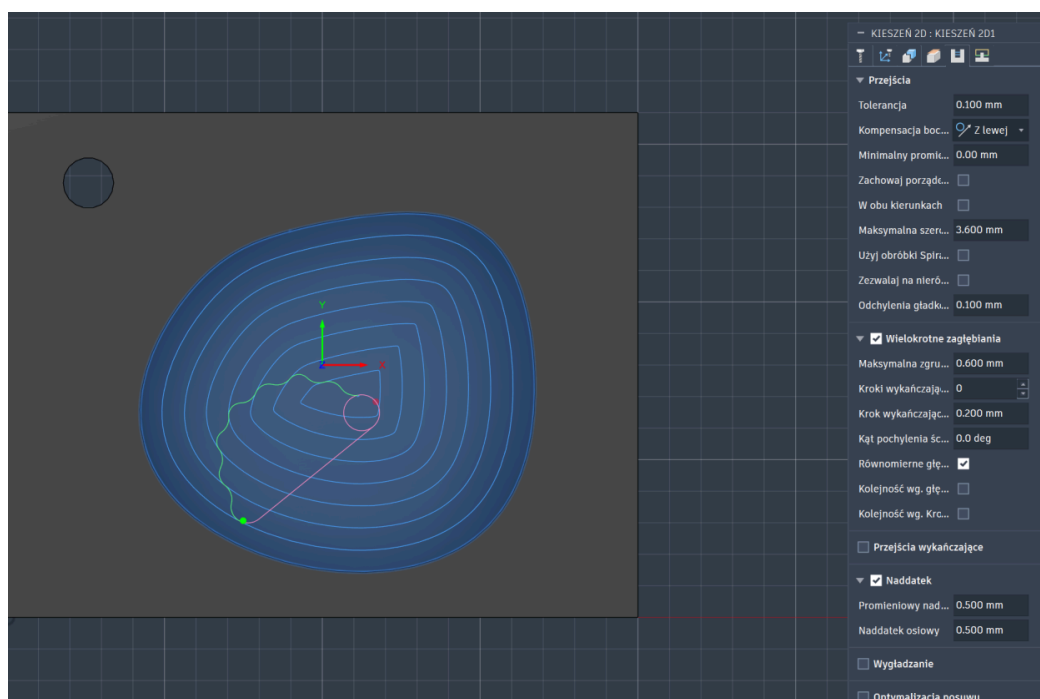


Fig. 4: Pocket milling

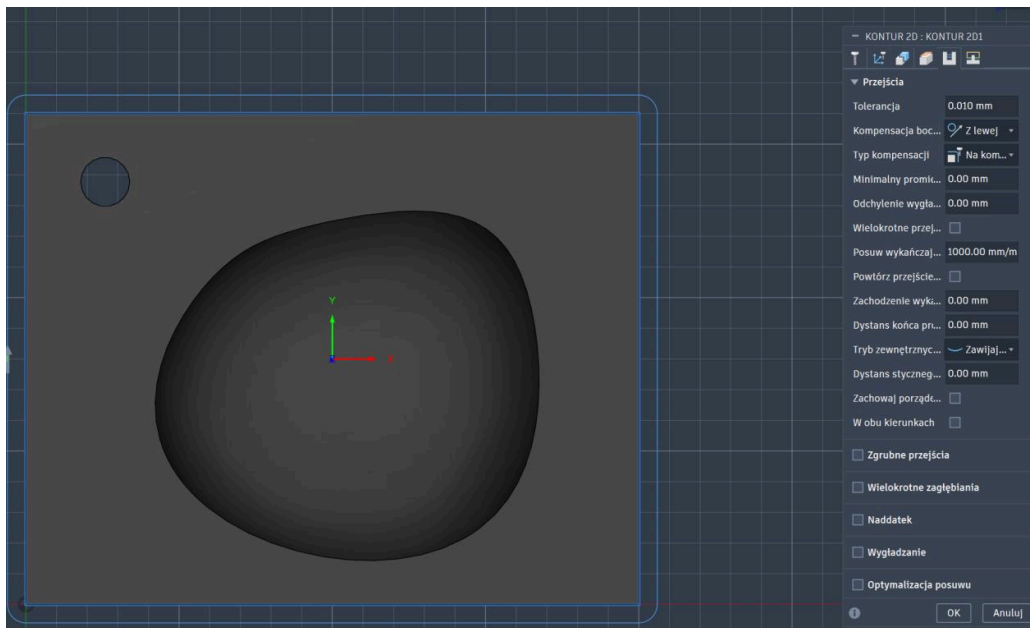


Fig. 5: Contour machining

7. G-code Program Analysis

7.1. Program Header

The program begins by setting the basic operating parameters:

```
N10 G90 G94 G17 G91.1
N15 G21
N20 G53 G0 Z0.
```

Meaning:

- G90 – absolute programming,
- G94 – feed in mm/min,
- G17 – XY plane,
- G21 – units in mm.

7.2. Face Milling

Facing was done with tool T1 (Ø10 mm).

```
(FACE2)
N25 T1 M6
N35 S5000 M3
N50 M8
```

The tool enters the material along an arc:

```
N75 G18 G3 X55.5 Z-1. I-1. K0. F1000.
```

This kind of entry reduces tool marks and improves surface quality.

7.3. Pocket Milling

The pocket was machined with tool T2 (Ø6 mm).

```
(2D POCKET1)  
N355 T2 M6
```

Multiple step-downs were used in the project:

```
N430 G3 X4.385 Y-9.742 Z2.2 I-1.844 J-2.173  
N435 X8.074 Y-5.396 Z1.9 I1.844 J2.173  
N440 X4.385 Y-9.742 Z1.6 I-1.844 J-2.173  
N445 X8.074 Y-5.396 Z1.3 I1.844 J2.173
```

The tool descended into the material gradually, which:

- reduces the load on the end mill,
- makes machining safer,
- limits the risk of tool damage.

Circular interpolation (G2/G3) was also used.

7.4. Contour Machining

The contour was machined with the Ø6 mm end mill. The program uses both linear and circular interpolation, which gives smooth tool motion.

7.5. Drilling

The hole was made with tool T3, a Ø8 mm drill.

The program uses the G81 drilling cycle:

```
G98 G81 X-37.056 Y28.855 Z-31. R4. F333.33
```

Parameter meanings:

- Z-31 – drilling depth,
- R4 – retract (clearance) plane,
- F333.33 – drilling feed rate.

After drilling, the cycle was cancelled with G80.

8. Project Requirements

The project requirements were met:

- face milling,
- pocket milling,
- contour machining,
- drilling with the G81 cycle,
- tool change,
- lead-in / lead-out into and out of the material,
- multiple step-downs,
- finishing pass,
- preparation of the process documentation.

9. Conclusions

The project made it possible to apply in practice the knowledge of CNC machine programming and of preparing a machining process in Autodesk Fusion. A complete machining program was developed, covering face milling, pocket milling, contour machining and drilling of a hole.

The most important element of the project was the use of multiple step-downs while milling the pocket. This allowed material to be removed gradually, which reduced the load on the tool and improved stability when machining steel. In addition, circular interpolation and arc entries gave smoother tool motion and reduced marks on the surface of the part.

Different types of tools were used, a tool change was performed during the program, and the G81 drilling cycle was applied. The program is correctly organised and contains the basic functions required in professional CNC machining.

The project provided experience in:

- analysing and interpreting G-code,
- selecting cutting parameters,
- planning the order of machining operations,
- preparing toolpaths,
- optimising the machining process.

The final program meets the requirements of the final project and allows the part to be machined correctly on a CNC machine tool.

10. G-code Program

The full G-code program is provided as a separate file: `gcode/1001.nc`.